

Roll No.

Y-7-C

Total No. of Questions : 31]

[Total No. of Printed Pages : 7

11th ARA(HZ)JKLEHUT2024

1407-C

MATHEMATICS

Time : 3 Hours]

[Maximum Marks : 80

Section-A

(Objective Type/Multiple Choice Questions)

1 each

1. Cardinal number of an empty set is :
(A) 0
(B) 1
(C) 2
(D) Infinite
2. Range of the function $f : \mathbb{R} \rightarrow \mathbb{R}$ given by $f(x) = |x - 1|$ is :
(A) $(-\infty, \infty)$
(B) $(-\infty, 0]$
(C) $[0, \infty)$
(D) None of these

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Turn Over

3. Maximum value of $\cos \theta$ is :
- (A) -1
(B) 0
(C) 1
(D) None of these
4. $i^{10} + i^{11}$ is equal to :
- (A) $-1 + i$
(B) $1 + i$
(C) $-1 - i$
(D) $1 - i$
5. If $n = 10$, $r = 3$, then value of nC_r is :
- (A) 100
(B) 120
(C) 110
(D) 520
6. Directrix of parabola $y^2 = -4x$ is
7. Eccentricity of the ellipse $\frac{x^2}{36} + \frac{y^2}{16} = 1$ is

(3)

8. If x is a natural number, then solution of $-12x > 30$ is $\{1, 2, 3, 4\}$.

(True/False)

9. Define a primary data.

10. If A and B are mutually exclusive events, then $P(A \text{ or } B) = P(A) + P(B)$

(True/False)

Section-B

(Very Short Answer Type Questions)

2 each

11. If $A = \{x \mid x \text{ is a two digit number such that sum of its digits is } 8\}$, find roaster form of A .

12. If $A = \{-1, 1\}$, find $A \times A \times A$.

13. Find the equation of line passing through the points $(-1, 1)$ and $(2, -4)$.

14. Find the equation of circle having centre $(0, 2)$ and radius 2.

15. Find the coordinates of the foci and vertices of the hyperbola

$$5y^2 - 9x^2 = 36.$$

(4)

16. Prove that :

$$\cos 2x = \frac{1 - \tan^2 x}{1 + \tan^2 x}$$

17. A die is thrown once, find the probability of "getting a number less than 6".
18. Find the variance of the data :

6, 7, 10, 12, 13, 4, 8, 12

19. If :

$$\frac{1}{6!} + \frac{1}{7!} = \frac{x}{8!},$$

find x .

20. Find the 20th term of the G.P. $\frac{5}{2}, \frac{5}{4}, \frac{5}{8}, \dots$

Section-C

(Short Answer Type Questions)

4 each

21. If $A = \{1, 2, 3, 5\}$ and $B = \{4, 6, 9\}$. Define a relation R from A to B by
- $$R = \{(x, y) \mid \text{the difference between } x \text{ and } y \text{ is odd; } x \in A, y \in B\}.$$
- Write R in roaster form and find domain and range of R .

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(5)

22. Prove that :

$$\cot x \cot 2x - \cot 2x \cot 3x - \cot 3x \cot x = 1$$

23. For any two complex numbers z_1 and z_2 , prove that :

$$\operatorname{Re}(z_1 z_2) = \operatorname{Re}(z_1) \operatorname{Re}(z_2) - \operatorname{Im}(z_1) \operatorname{Im}(z_2)$$

24. Solve the inequality :

$$\frac{(2x-1)}{3} \geq \frac{(3x-2)}{4} - \frac{(2-x)}{5}$$

25. Find the equation of the line perpendicular to the line $x - 7y + 5 = 0$ and having x-intercept 3. <https://www.jkbosenonline.com>

26. Evaluate :

$$\lim_{x \rightarrow 3} \frac{x^4 - 81}{2x^2 - 5x - 3}$$

27. If :

$$y = \frac{4x + 5 \sin x}{3x + 7 \cos x},$$

find $\frac{dy}{dx}$

28. In how many ways can one select a cricket team of 11 from 17 players in which only 5 players can bowl if each cricket team of 11 must include exactly 4 bowlers ?

Section-D

(Long Answer Type Questions)

6 each

29. Find ;

$$\sin \frac{x}{2}, \cos \frac{x}{2} \text{ and } \tan \frac{x}{2}$$

if $\tan x = -\frac{4}{3}$, x is in quadrant 2nd.

Or

Prove that :

$$\sin x + \sin 3x + \sin 5x + \sin 7x = 4 \cos x \cos 2x \sin 4x$$

30. If $\frac{a^n + b^n}{a^{n-1} + b^{n-1}}$ is the A.M. between a and b , then find the value of n .

Or

Find the sum to n terms of the sequence 8, 88, 888, 8888,

31. Find the mean deviation about the median for the following data :

Marks	No. of Girls
0—10	6
10—20	8
20—30	14
30—40	16
40—50	4
50—60	2

(7)

Or

A committee of two persons is selected from two men and two women.

What is the probability that the committee will have :

- (a) No man
- (b) One man
- (c) Two men

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